

This has enabled Alphabet Energy to design and build its products in small batches.

By contrast, most successful clean technology startups have had to make substantial early investments. The solar panel provider SunPower had to build in factories to manufacture its panels and partner with installers before becoming fully operational. Similarly, BrightSource raised \$291 million to build and operate large-scale solar plants without delivering a watt to a single customer.

Instead of having to invest time and money in expensive fabrication facilities, Alphabet is able to take advantage of the massive existing infrastructure that produces silicon wafers for computer electronics. As a result, Alphabet can go from a product concept to holding a physical version in its hand in just six weeks from end to end. Alphabet's challenge has been to find the combination of performance, price, and physical shape that is a match for early customers. Although its technology has revolutionary potential, early adopters will deploy it only if they can see a clear return on investment.

It might seem that the most obvious market for Alphabet's technology would be power plants, and indeed, that was the team's initial hypothesis. Alphabet hypothesized that simple cycle gas turbines would be an ideal application; these turbines, which are similar to jet engines strapped to the ground, are used by power generators to provide energy for peak demand. Alphabet believed that attaching its semiconductors to those turbines would be simple and cheap.

The company went about testing this hypothesis in small batches by building small-scale solutions for its customers as a way of learning. As with many initial ideas, their hypothesis was disproved quickly. Power companies have a low tolerance for risk, making them unlikely to become early adopters. Because it wasn't weighed down by a large-batch approach, Alphabet was ready to pivot after just three months of investigation.

Alphabet has eliminated many other potential markets as well, leading to a series of customer segment pivots. The company's current efforts are focused on manufacturing firms, which have the